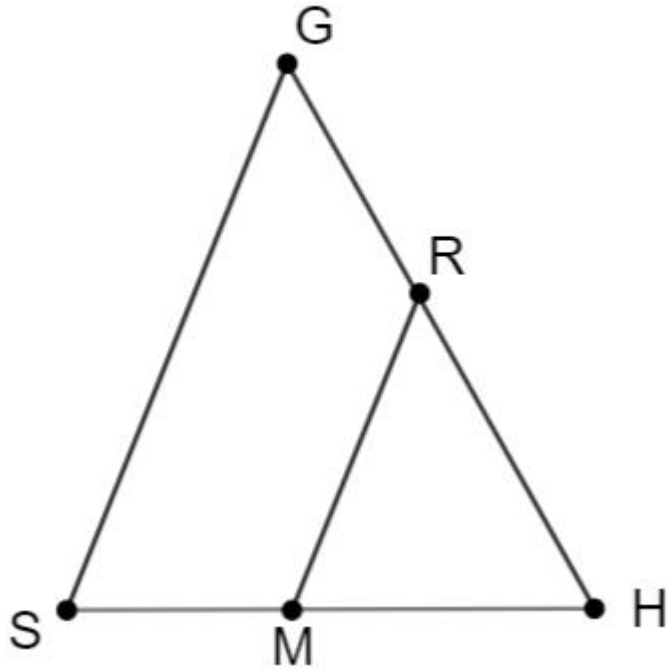


6.9.2 Activity

Proof A

Given: $\overline{GS} \parallel \overline{RM}$

Prove: $\triangle RHM \sim \triangle GHS$



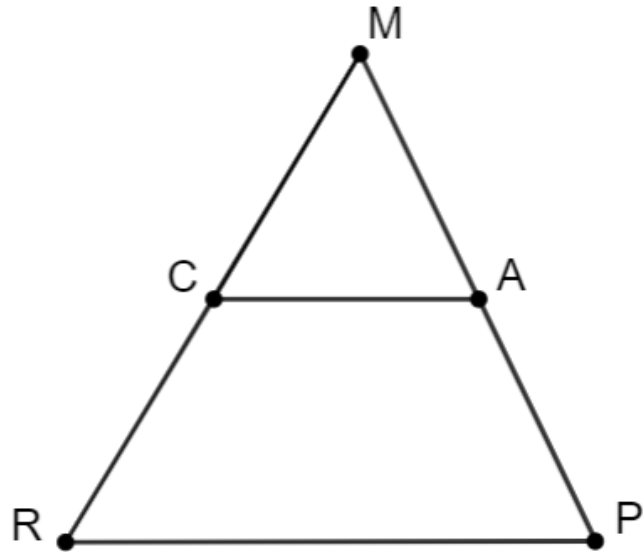
Statement	Reason
$\overline{GS} \parallel \overline{RM}$	Given
$\angle H \cong \angle H$	Reflexive Property of Congruence
$\angle GSH \cong \angle RMH$	If two parallel lines are intersected by a transversal, corresponding angles are congruent
$\triangle RHM \sim \triangle GHS$	Angle-Angle Similarity Theorem

Proof B

Given: $MC = \frac{2}{5}MR$ and

$$MA = \frac{2}{5}MP$$

Prove: $\Delta MCA \sim \Delta MRP$

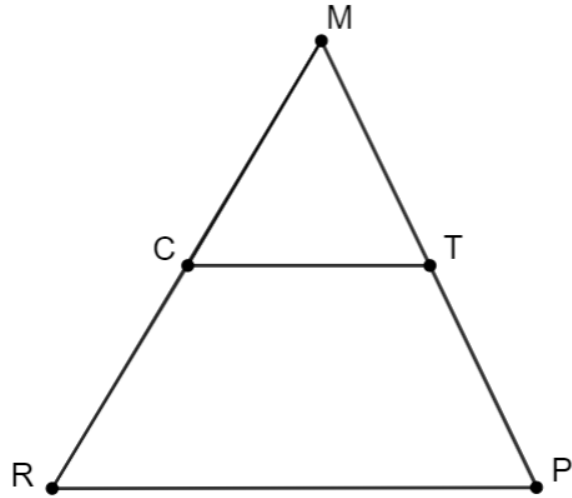


Statement	Reason
$MC = \frac{2}{5}MR$ and $MA = \frac{2}{5}MP$	Given
$\frac{MC}{MR} = \frac{2}{5}, \frac{MA}{MP} = \frac{2}{5}$	Division Property of Equality
$\frac{MC}{MR} = \frac{MA}{MP}$	Substitution Property
$\angle M \cong \angle M$	Reflexive Property of Congruence
$\Delta MCA \sim \Delta MRP$	Side-Angle-Side Similarity Theorem

Proof C

Given: $\angle RPT$ and $\angle CTP$ are supplementary

Prove: $\triangle MCT \sim \triangle MRP$

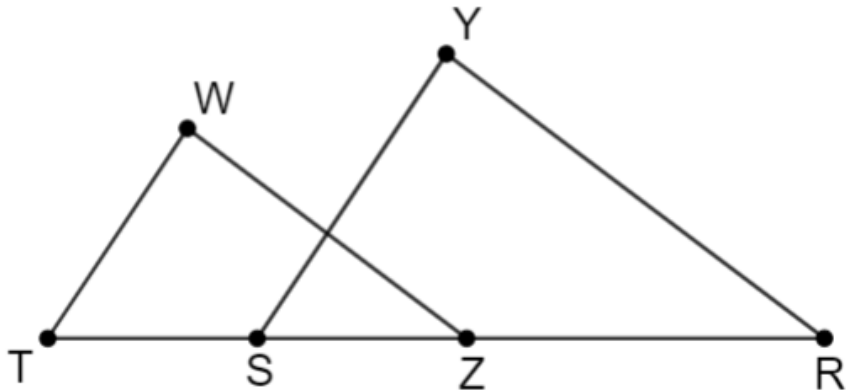


Statement	Reason
$\angle RPT$ and $\angle CTP$ are supplementary	Given
$m\angle RPT + m\angle CTP = 180^\circ$	Definition of Supplementary Angles
$\angle CTM$ and $\angle CTP$ are supplementary	If two angles form a linear pair, they are supplementary
$m\angle CTM + m\angle CTP = 180^\circ$	Definition of Supplementary Angles
$m\angle CTM + m\angle CTP = m\angle RPT + m\angle CTP$	Transitive Property of Equality
$m\angle CTM = m\angle RPT$	Subtraction Property of Equality
$\angle CTM \cong \angle RPT$	Definition of Congruence
$\angle M \cong \angle M$	Reflexive Property of Congruence
$\triangle MCT \sim \triangle MRP$	Angle-Angle Similarity Theorem

Proof D

Given: $\overline{WT} \parallel \overline{YS}$,
 $\overline{WZ} \parallel \overline{YR}$

Prove:
 $\triangle WZT \sim \triangle YRS$

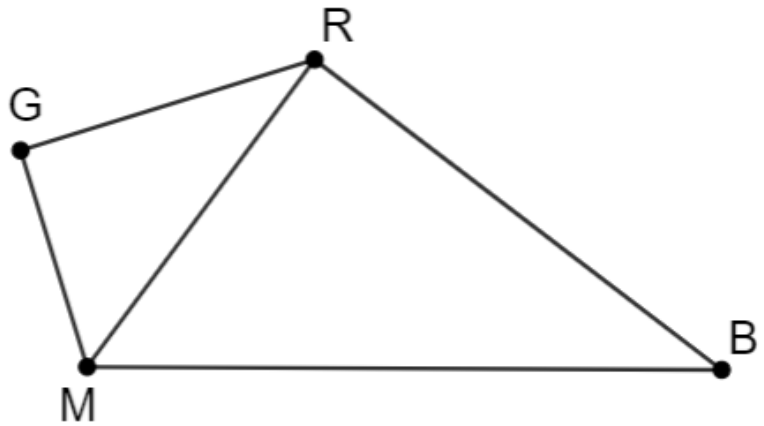


Statement	Reason
$\overline{WT} \parallel \overline{YS}$	Given
$\angle WZT \cong \angle YSR$	If two parallel lines are intersected by a transversal, corresponding angles are congruent
$\overline{WZ} \parallel \overline{YR}$	Given
$\angle WZT \cong \angle YRS$	If two parallel lines are intersected by a transversal, corresponding angles are congruent
$\triangle WZT \sim \triangle YRS$	Angle-Angle Similarity Theorem

Proof E

Given: $GM = \frac{3}{5}RM$,
 $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$

Prove:
 $\Delta MGR \sim \Delta MRB$

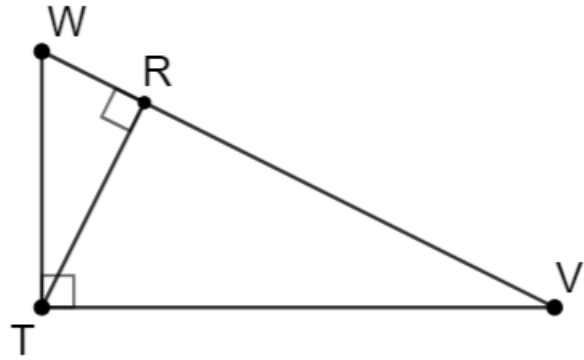


Statement	Reason
$GM = \frac{3}{5}RM$, $RG = \frac{3}{5}BR$ $RM = \frac{3}{5}BM$	Given
$\frac{GM}{RM} = \frac{3}{5}$, $\frac{RG}{BR} = \frac{3}{5}$, $\frac{RM}{BM} = \frac{3}{5}$	Division Property of Equality
$\frac{GM}{RM} = \frac{RG}{BR} = \frac{RM}{BM}$	Substitution Property
$\Delta MGR \sim \Delta MRB$	Side-Side-Side Similarity Theorem

Proof F

Given: $\overline{WT} \perp \overline{VT}$ and
 $\overline{WR} \perp \overline{RT}$

Prove: $\triangle RTW \sim \triangle TVW$

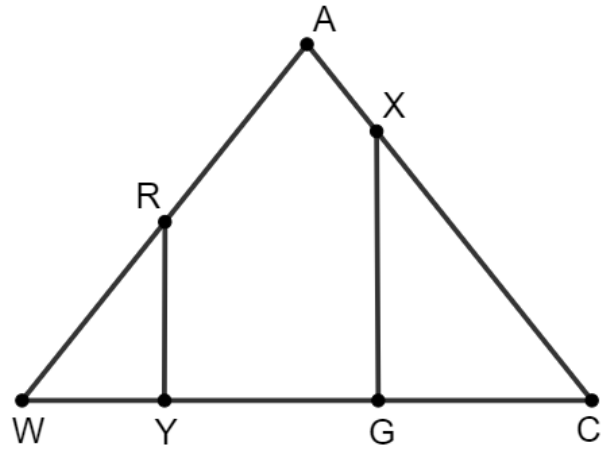


Statement	Reason
$\overline{WT} \perp \overline{VT}$	Given
$\angle VTW$ is a right angle	Definition of Perpendicular Lines
$m\angle VTW = 90^\circ$	Definition of Right Angle
$\overline{WR} \perp \overline{RT}$	Given
$\angle TRW$ is a right angle	Definition of Perpendicular Lines
$m\angle TRW = 90^\circ$	Definition of Right Angle
$m\angle TRW = m\angle VTW$	Substitution
$\angle TRW \cong \angle VTW$	Definition of Congruence
$\angle W \cong \angle W$	Reflexive Property of Congruence
$\triangle RTW \sim \triangle TVW$	Angle-Angle Similarity Theorem

Proof G

Given: $\overline{WA} \cong \overline{CA}$, $\overline{RY} \perp \overline{WC}$,
 $\overline{XG} \perp \overline{WC}$

Prove: $\Delta RYW \sim \Delta XGC$



Statement	Reason
$\overline{WA} \cong \overline{CA}$	Given
$WA = CA$	Definition of Congruence
ΔWAC is Isosceles	Definition of Isosceles Triangles
$m\angle AWC = m\angle ACW$	Definition of Isosceles Triangles
$\angle AWC \cong \angle ACW$	Definition of Congruence
$\overline{RY} \perp \overline{WC}$	Given
$\angle RYW$ is a right angle	Definition of Perpendicular Lines
$m\angle RYW = 90^\circ$	Definition of Right Angle
$\overline{XG} \perp \overline{WC}$	Given
$\angle XGC$ is a right angle	Definition of Perpendicular Lines
$m\angle XGV = 90^\circ$	Definition of Right Angle
$m\angle RYW = m\angle XGV$	Substitution Property
$\angle RYW \cong \angle XGV$	Definition of Congruence
$\Delta RYW \sim \Delta XGC$	Angle-Angle Similarity Theorem